



Telco Customer Churn Prediction using Machine Learning

An End-to-End Data Science Case Study with Python

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Role: Data Science & Machine Learning Project

Project Type: End-to-End Machine Learning Case Study

Tools & Technologies:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Joblib

Project Domain: Telecommunications | Customer Analytics | Predictive Analytics

Executive Summary

1.1 Introduction

Customer retention has become one of the most important priorities for telecom companies operating in highly competitive markets. While acquiring new customers often requires significant marketing and operational investment, retaining existing customers is generally more cost-effective and contributes directly to long-term business growth. Even a small increase in customer churn can lead to substantial revenue loss, making early identification of at-risk customers a valuable business capability.

This project presents an end-to-end machine learning solution designed to predict customer churn using historical customer information. By analyzing demographic details, account information, service subscriptions, billing patterns, and customer behavior, the model identifies customers who are more likely to discontinue their services. These predictions can help organizations implement proactive retention strategies, optimize marketing efforts, and improve overall customer satisfaction.

1.2 Project Objective

The primary objective of this project was to develop a predictive machine learning model capable of identifying customers who are likely to churn based on their historical data. In addition to building an accurate model, the project focused on understanding the factors influencing churn and translating technical findings into actionable business recommendations.

1.2 Business Objectives

The project was designed to address the following business goals:

- Predict customers who are at risk of leaving the telecom service.
- Identify the key factors contributing to customer churn.
- Support customer retention initiatives through data-driven insights.
- Enable proactive decision-making using predictive analytics.

- Reduce revenue loss associated with customer attrition.
- Demonstrate how machine learning can solve a real-world business problem

1.4 Project Workflow

The project followed a structured Data Science lifecycle:

Business Understanding → Data Collection → Data Cleaning & Preprocessing → Exploratory Data Analysis → Feature Engineering → Model Development → Model Evaluation → Hyperparameter Tuning → Business Insights → Recommendations

1.5 Technologies Used

Category	Technologies
Programming Language	Python
Development Environment	Google Colab
Data Manipulation	Pandas, NumPy
Data Visualization	Matplotlib
Machine Learning	Scikit-learn
Model Serialization	Joblib

1.6 Machine Learning Models Evaluated

To determine the most suitable **predictive model**, multiple supervised machine learning algorithms were trained and evaluated.

The models included:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Each model was assessed using appropriate evaluation metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC Score. Hyperparameter tuning was then performed to further improve model performance and identify the most effective solution for the business problem.

1.7 Key Outcomes

The project successfully demonstrated that machine learning can be applied to predict customer churn with strong predictive performance while also generating meaningful business insights. Beyond building a classification model, the project highlighted important customer characteristics associated with churn and translated those findings into practical recommendations that could support customer retention initiatives.

The project also reinforced the importance of combining technical implementation with business understanding, showing how predictive analytics can help organizations make more informed, data-driven decisions.

Project at a Glance

2.1 Project Overview :

Attribute	Details
Project Name	Telco Customer Churn Prediction using Machine Learning
Project Type	End-to-End Machine Learning Classification Project
Domain	Telecommunications

Business Function	Customer Retention & Predictive Analytics
Problem Type	Binary Classification
Target Variable	Churn (Yes / No)
Programming Language	Python
Development Environment	Google Colab
Machine Learning Library	Scikit-learn
Project Status	Completed

2.2 Business Objective:

To develop a machine learning solution capable of predicting whether a telecom customer is likely to discontinue the service. The project aims to help businesses identify at-risk customers early and support proactive customer retention strategies through data-driven decision-making.

2.3 Dataset Snapshot:

Telco Customer Churn Data

Dataset Information	Value
Total Records	7,043
Total Features	21
Target Variable	Churn
Numerical Features	3
Categorical Features	18
Missing Values	Handled during preprocessing
Duplicate Records	Checked and removed

2.4 Technology Stack:

Category	Tools Used
Programming	Python
Notebook	Google Colab
Data Analysis	Panda, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Model Storage	Joblib

2.5 Machine Learning Models Evaluated :

Model	Purpose
Logistic Regression	Baseline Classification Model
Decision Tree	Rule-Based classification
Random Forest	Ensemble learning
K-Nearest Neighbors (KNN)	Distance-based classification
Support Vector Machine (SVM)	Margin-based classification

2.6 Final Selected Model :

Metric	Result
Selected Model	Logistic Regression
Accuracy	80.62%

ROC-AUC Score	84.22%
Business Focus	Customer Retention

About the Project

3.1 Introduction

In today's highly competitive telecommunications industry, retaining existing customers has become just as important as acquiring new ones. Telecom companies invest significant resources in attracting new subscribers through marketing campaigns, promotional offers, and service expansions. However, customer acquisition is often far more expensive than customer retention. Losing existing customers not only results in immediate revenue loss but can also negatively impact long-term profitability, customer lifetime value, and brand reputation.

With the growing availability of customer data, organizations are increasingly adopting Data Science and Machine Learning techniques to better understand customer behavior and predict future outcomes. One of the most valuable applications of predictive analytics in the telecom sector is **Customer Churn Prediction**, which helps businesses identify customers who are at risk of discontinuing their services before they actually leave.

This project demonstrates how machine learning can be used to analyze customer demographics, subscription details, service usage, billing patterns, and account information to predict the likelihood of customer churn. By identifying high-risk customers in advance, businesses can implement targeted retention strategies, improve customer satisfaction, and reduce avoidable revenue loss.

3.2 Business Problem Statement

A leading telecommunications company has experienced a steady increase in customer churn over recent months. Although the company continues to attract new customers, a

significant number of existing subscribers are discontinuing their services, resulting in reduced recurring revenue and increased customer acquisition costs.

The management team has identified customer retention as a strategic priority and wants to adopt a data-driven approach to understand why customers are leaving. Instead of relying solely on historical reports, the organization aims to build a predictive system capable of identifying customers who are likely to churn before they make the decision to leave.

As a **Data Scientist**, the objective is to analyze customer data, identify the factors contributing to churn, develop a reliable machine learning model, and provide actionable business recommendations that support customer retention initiatives.

3.3 Project Objectives

The project was undertaken with the following objectives:

I) Primary Objectives

- Develop a machine learning model capable of predicting customer churn.
- Identify the key factors influencing customer attrition.
- Compare multiple machine learning algorithms to determine the most suitable model.
- Optimize model performance through hyperparameter tuning.
- Evaluate the model using industry-standard performance metrics.

II) Business Objectives

- Enable proactive customer retention strategies.
- Reduce revenue loss caused by customer churn.
- Improve customer satisfaction through targeted interventions.
- Support business decision-making using predictive analytics.
- Demonstrate the practical application of **Data Science** in solving real-world business problems.

3.4 Scope of the Project

The scope of this project includes the complete **Data Science** lifecycle, beginning with data understanding and ending with business recommendations.

The project covers:

- Data collection and exploration
- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Feature engineering
- Model development
- Model evaluation
- Hyperparameter tuning
- Feature importance analysis
- Business insights and recommendations
- Deployment preparation

While the current project focuses on model development and deployment planning, future enhancements may include cloud deployment, real-time prediction services, **Explainable AI (XAI)**, and continuous model monitoring.

Real-World Business Scenario

4.1 Business Scenario

A leading telecommunications company has observed a steady increase in customer churn over the past few months, resulting in declining recurring revenue and rising customer acquisition costs. Although the company continues to attract new subscribers, retaining existing customers has become a significant business challenge.

The management team wants to adopt a **data-driven approach** to identify customers who are likely to discontinue their service before they leave. By predicting customer churn in advance, the company aims to implement targeted retention strategies, improve customer satisfaction, and optimize marketing investments.

As a **Data Scientist**, my responsibility was to analyze historical customer data, identify the key factors influencing churn, build a predictive **Machine Learning** model, and provide actionable business recommendations to support customer retention.

4.2 Business Problem

The company lacks a predictive system to identify customers at risk of churn. As a result, retention efforts are often reactive rather than proactive, leading to unnecessary customer loss and increased business costs.

The challenge was to develop a reliable classification model capable of predicting whether a customer is likely to churn based on demographic information, service usage, account details, and billing history.

4.3 Project Goals

The project was designed to achieve the following objectives:

- Develop an accurate **Machine Learning** model to predict customer churn.
- Identify the most influential factors contributing to customer attrition.
- Compare multiple classification algorithms to determine the best-performing model.
- Generate actionable insights to support customer retention strategies.
- Demonstrate an end-to-end **Data Science** workflow, from data preparation to business recommendations.

4.3 Project Workflow

Business Understanding → Data Collection → Data Cleaning & Preprocessing → Exploratory Data Analysis (EDA) → Feature Engineering → Model Development → Model Evaluation → Hyperparameter Tuning → Business Insights & Recommendations

4.4 Success Criteria

The project was considered successful if it could:

- Predict customer churn with strong and reliable performance.
- Identify high-risk customers before they leave.
- Provide meaningful business insights from customer data.
- Support proactive customer retention through data-driven recommendations.

- Deliver a solution that can be extended for real-world deployment.

4.5 Business Value

The project demonstrates how **Predictive Analytics** can help organizations transform historical customer data into actionable business intelligence. By identifying customers with a high likelihood of churn, businesses can focus retention efforts more effectively, reduce revenue loss, and improve long-term customer relationships.

Data Preprocessing & Data Cleaning

5.1 Objective

The objective of this phase was to prepare the raw dataset for **machine learning** by identifying data quality issues, handling missing values, correcting data types, checking duplicate records, and removing unnecessary features. A clean and reliable dataset is essential for building an accurate and trustworthy predictive model.

5.2 Dataset Loading & Initial Exploration

The dataset was first imported into Google Colab using the Pandas library. An initial inspection was performed to understand the structure of the dataset, identify feature types, and verify the overall quality of the data.

Key Python Code:

```
df = pd.read_csv('/content/drive/MyDrive/Data Science Portfolio Website (Souparna)/Projects/Data Science , AI ML , GEN AI Project/AI ML/Telco Customer
```

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProt
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                7043 non-null   object
2   SeniorCitizen         7043 non-null   int64
3   Partner               7043 non-null   object
4   Dependents            7043 non-null   object
5   tenure               7043 non-null   int64
6   PhoneService          7043 non-null   object
7   MultipleLines         7043 non-null   object
8   InternetService       7043 non-null   object
9   OnlineSecurity        7043 non-null   object
10  OnlineBackup          7043 non-null   object
11  DeviceProtection      7043 non-null   object
12  TechSupport           7043 non-null   object
```

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

Business Interpretation

The initial exploration confirmed that the dataset contained customer demographics information, subscribed services, account details, billing information, and the target variable (Churn). This step also helped identify columns requiring further preprocessing before model development.

5.3 Missing Value Analysis

A detailed assessment was conducted to identify missing values across all features. During this analysis, the **Total Charges** column appeared as an **Object** data type because blank values were stored as empty strings instead of numeric values.

Key Python Code:

```
df.isnull().sum()
```

	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0

```
missing_percentage = (df.isnull().sum() / len(df)) * 100
```

	0
customerID	0.0
gender	0.0
SeniorCitizen	0.0
Partner	0.0
Dependents	0.0
tenure	0.0
PhoneService	0.0
MultipleLines	0.0
InternetService	0.0
OnlineSecurity	0.0

Business Interpretation

The missing values were associated with customers having a **tenure of 0 months**, indicating newly joined customers who had not yet generated any total billing. Converting the column to a numeric format ensured accurate calculations during later stages of analysis.

5.4 Data Cleaning

After identifying data quality issues, appropriate preprocessing techniques were applied to prepare the dataset for machine learning.

The following steps were completed:

- Converted **TotalCharges** to numeric format.
- Handled missing values in the **TotalCharges** column.
- Verified duplicate records.
- Removed unnecessary identifier columns (**customerID**).
- Validated the dataset after cleaning.

Key Python Code:

```
df_clean.head()
```

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	Tech
0	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	
1	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	
2	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	
3	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes	
4	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	

Business Interpretation

Removing the identifier field eliminated attributes that carried no predictive value, while validating the dataset ensured that the remaining features were suitable for machine learning. These preprocessing steps improved data quality and reduced the likelihood of biased or misleading model predictions.

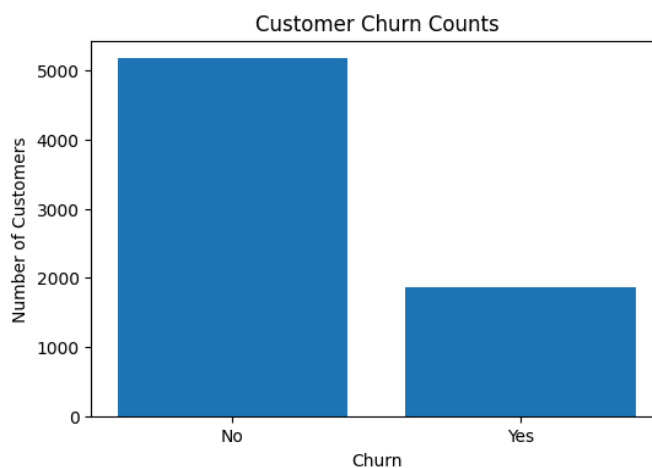
Exploratory Data Analysis (EDA)

6.1 Objective

The objective of **Exploratory Data Analysis (EDA)** was to understand customer behavior, identify churn patterns, and uncover insights to improve model performance and support data-driven business decisions.

6.2 Target Variable Analysis

The first step was to analyze the distribution of the target variable (**Churn**) to understand the proportion of customers who stayed versus those who left the telecom service.



Business Insight

The analysis revealed that most customers remained with the company, while a smaller but significant group had churned. This indicates an **imbalanced dataset**, making evaluation metrics such as **Precision, Recall, F1-Score, and ROC-AUC** more reliable than accuracy alone.

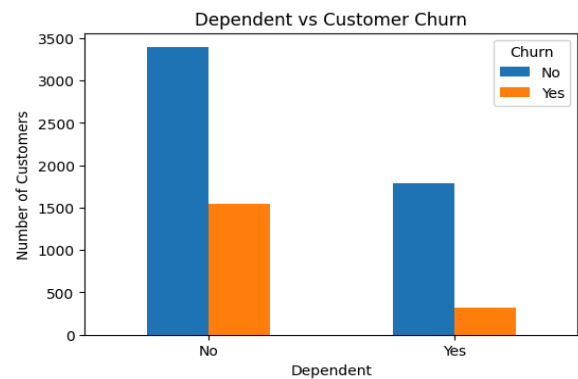
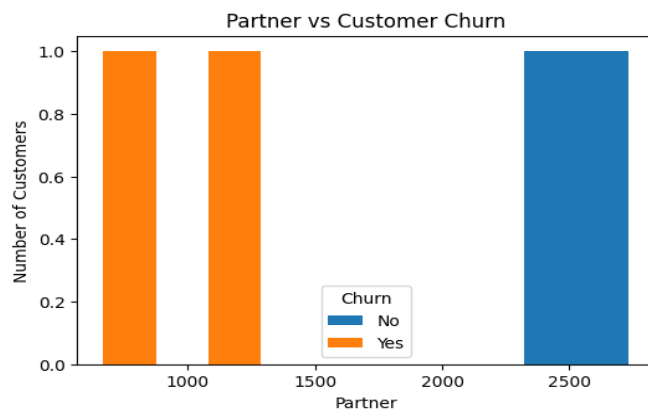
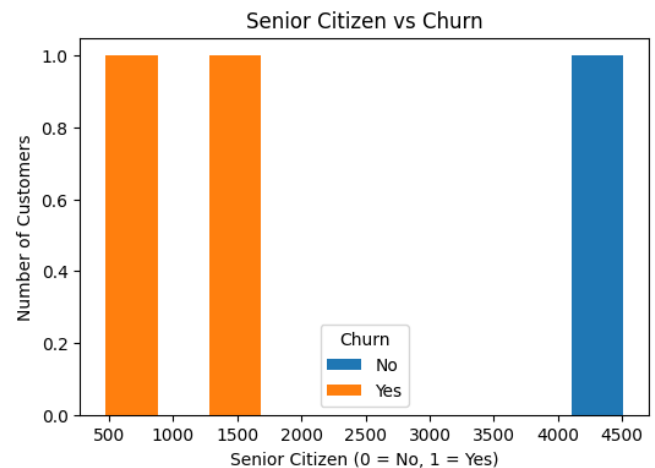
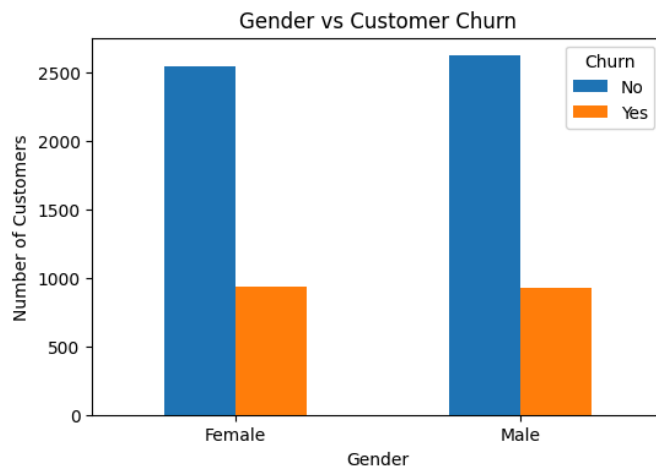
6.3 Customer Demographics

Customer demographic features were analyzed to understand how different customer groups influence churn.

The following features were explored:

- Gender
- Senior Citizen

- Partner
- Dependents



Business Insight

Senior citizens and customers without partners and dependents showed relatively higher churn rates, suggesting that customer demographics can influence retention and should be considered while designing targeted engagement strategies.

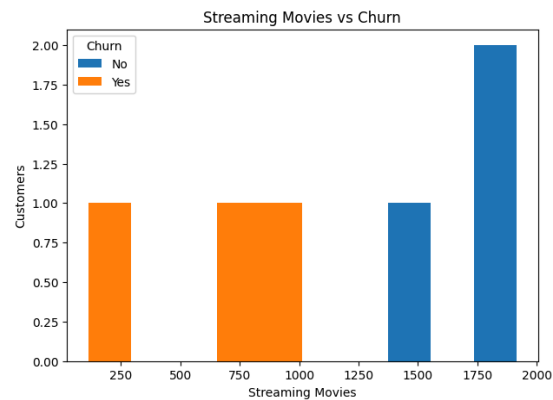
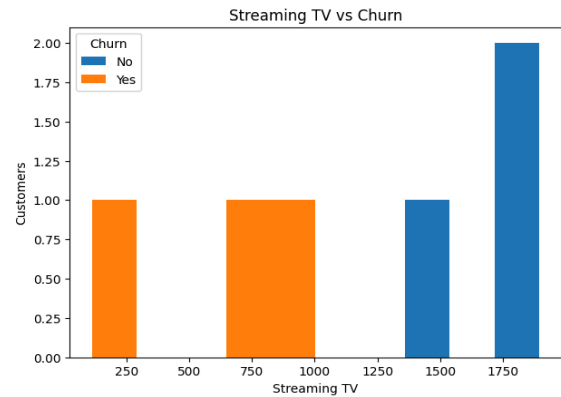
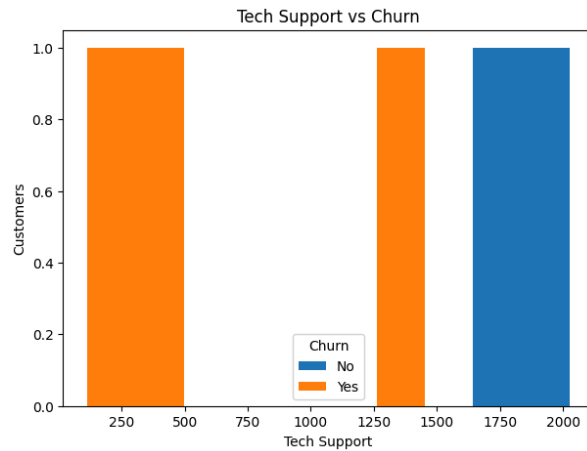
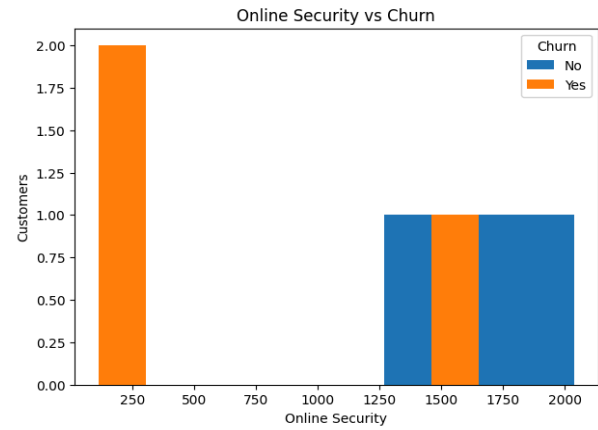
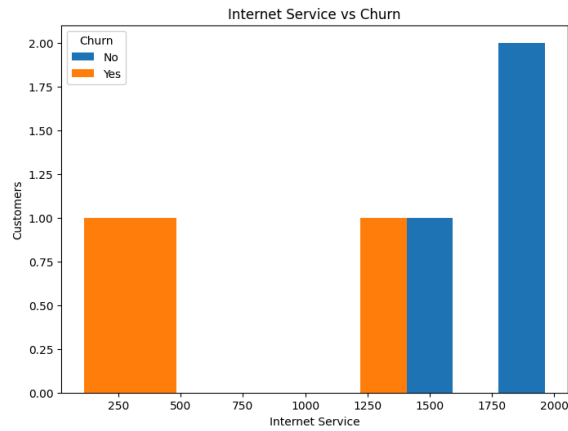
6.4 Service Usage Analysis

The impact of subscribed telecom services on customer churn was evaluated.

The following services were analyzed:

- Internet Service
- Online Security

- Tech Support
- Streaming TV
- Streaming Movies



Business Insight

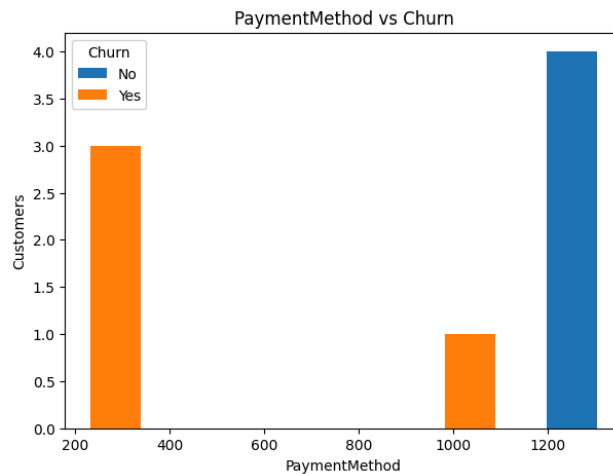
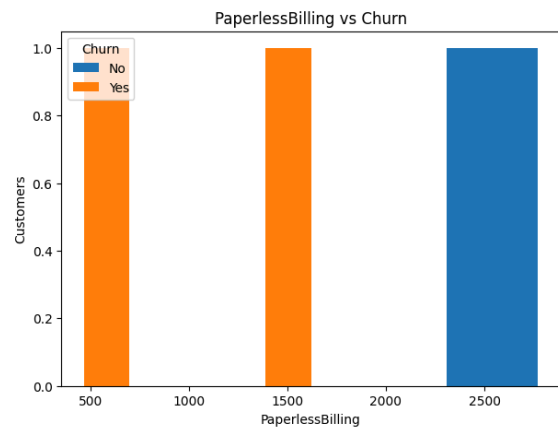
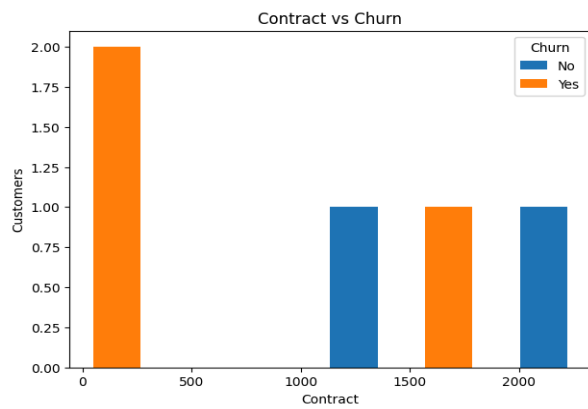
Customers without value-added services such as **Online Security** and **Tech Support** exhibited higher churn rates. These services present opportunities for customer retention through personalized offers and bundled plans.

6.4 Contract & Payment Analysis

Customer churn was analyzed across different contract and payment methods.

The following features were evaluated:

- Contract
- Paperless Billing
- Payment Method



Business Insight

Customers on **Month-to-Month contracts** and those using **Electronic Cheque** as their payment method demonstrated a higher likelihood of churn. Promoting long-term contracts and encouraging alternative payment methods may help improve customer retention.

Feature Engineering & Machine Learning Model Development

7.1 Objective

The objective of this phase was to transform the cleaned dataset into a machine learning-ready format, train multiple classification models, and identify the most suitable model for predicting customer churn.

7.2 Feature Engineering

Categorical variables were converted into numerical format so that the machine learning algorithms could process them. The target variable, **Churn**, was also mapped into binary values.

Key Python Code

```
df_clean.columns

Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
       'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
       'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
       'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',
       'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

```
#Seperate the target variable
```

```
X = df_clean.drop("Churn", axis=1)
y = df_clean["Churn"]
```

Business Interpretation

Encoding converted customer categories such as contract type, payment method, and internet service into model-readable features while preserving the information required to identify churn patterns.

7.3 Feature Scaling

Numerical variables with different value ranges were standardized to ensure that distance-based and margin-based algorithms were not influenced by scale differences.

Key Python Code

Module 5 - Feature Scaling

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

numerical_columns = [
    "tenure",
    "MonthlyCharges",
    "TotalCharges"
]

X[numerical_columns] = scaler.fit_transform(
    X[numerical_columns]
)
```

Business Interpretation

Feature scaling created a consistent numerical range across tenure and billing-related variables, improving the reliability of models such as **KNN, Logistic Regression, and SVM**.

7.4 Train-Test Split

The dataset was divided into training and testing sets. Stratification was used to maintain a similar churn distribution in both datasets.

Key Python Code

Module 6 - Train-Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

Business Interpretation

The training dataset was used to learn customer churn patterns, while the testing dataset measured how well the models could predict outcomes for unseen customers.

7.5 Model Development

Five supervised machine learning algorithms were trained and evaluated:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors
- Support Vector Machine

Key Python Code

Import Algorithm

```
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC

from sklearn.metrics import accuracy_score
```

Logistic Regression (Baseline Model)

```
lr_model = LogisticRegression(max_iter=1000)

lr_model.fit(X_train, y_train)

y_pred_lr = lr_model.predict(X_test)

lr_accuracy = accuracy_score(y_test, y_pred_lr)

print("Logistic Regression Accuracy:", round(lr_accuracy, 4))
```

Logistic Regression Accuracy: 0.8062

```
print(f"Logistic Regression Accuracy: {lr_accuracy * 100:.2f}%")
```

Logistic Regression Accuracy: 80.62%

Decision Tree

```
dt_model = DecisionTreeClassifier(
    random_state=42,
    max_depth=5
)

dt_model.fit(X_train, y_train)

y_pred_dt = dt_model.predict(X_test)

dt_accuracy = accuracy_score(y_test, y_pred_dt)

print("Decision Tree Accuracy:", round(dt_accuracy, 4))
```

Decision Tree Accuracy: 0.7942

```
print(f"Decision Tree Accuracy: {dt_accuracy * 100:.2f}%")
```

Decision Tree Accuracy: 79.42%

Random Forest

```
rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42,
    max_depth=10
)

rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)

rf_accuracy = accuracy_score(y_test, y_pred_rf)

print("Random Forest Accuracy:", round(rf_accuracy, 4))
```

Random Forest Accuracy: 0.8041

```
print(f"Random Forest Accuracy: {rf_accuracy * 100:.2f}%")
```

Random Forest Accuracy: 80.41%

K-Nearest Neighbors (KNN)

```
knn_model = KNeighborsClassifier(n_neighbors=5)

knn_model.fit(X_train, y_train)

y_pred_knn = knn_model.predict(X_test)

knn_accuracy = accuracy_score(y_test, y_pred_knn)

print("KNN Accuracy:", round(knn_accuracy, 4))
```

KNN Accuracy: 0.763

```
print(f"KNN Accuracy: {knn_accuracy * 100:.2f}%")
```

KNN Accuracy: 76.30%

Support Vector Machine (SVM)

```
svm_model = SVC(kernel='rbf')
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict(X_test)
svm_accuracy = accuracy_score(y_test, y_pred_svm)
print("SVM Accuracy:", round(svm_accuracy, 4))

SVM Accuracy: 0.7949

print(f"SVM Accuracy: {svm_accuracy * 100:.2f}%")

SVM Accuracy: 79.49%
```

7.6 Model Performance Comparison

Machine Learning Model	Accuracy
Logistic Regression	80.62%
Decision Tree	79.42%
Random Forest	80.41%
K-Nearest Neighbors	76.30%
Support Vector Machine	79.49%

Key Observation

Logistic Regression achieved the highest overall accuracy and provided a strong balance between predictive performance and model interpretability.

7.7 Final Model Selection

Logistic Regression was selected as the primary model because it:

- Achieved the highest test accuracy
- Produced a strong ROC-AUC score
- Offered clear and interpretable feature coefficients
- Supported probability-based churn prediction
- Was suitable for threshold tuning and customer risk segmentation

Business Interpretation

For the customer retention use case, model interpretability is especially valuable. Logistic Regression allows business teams to understand which customer characteristics increase or decrease churn risk rather than relying only on a final prediction.

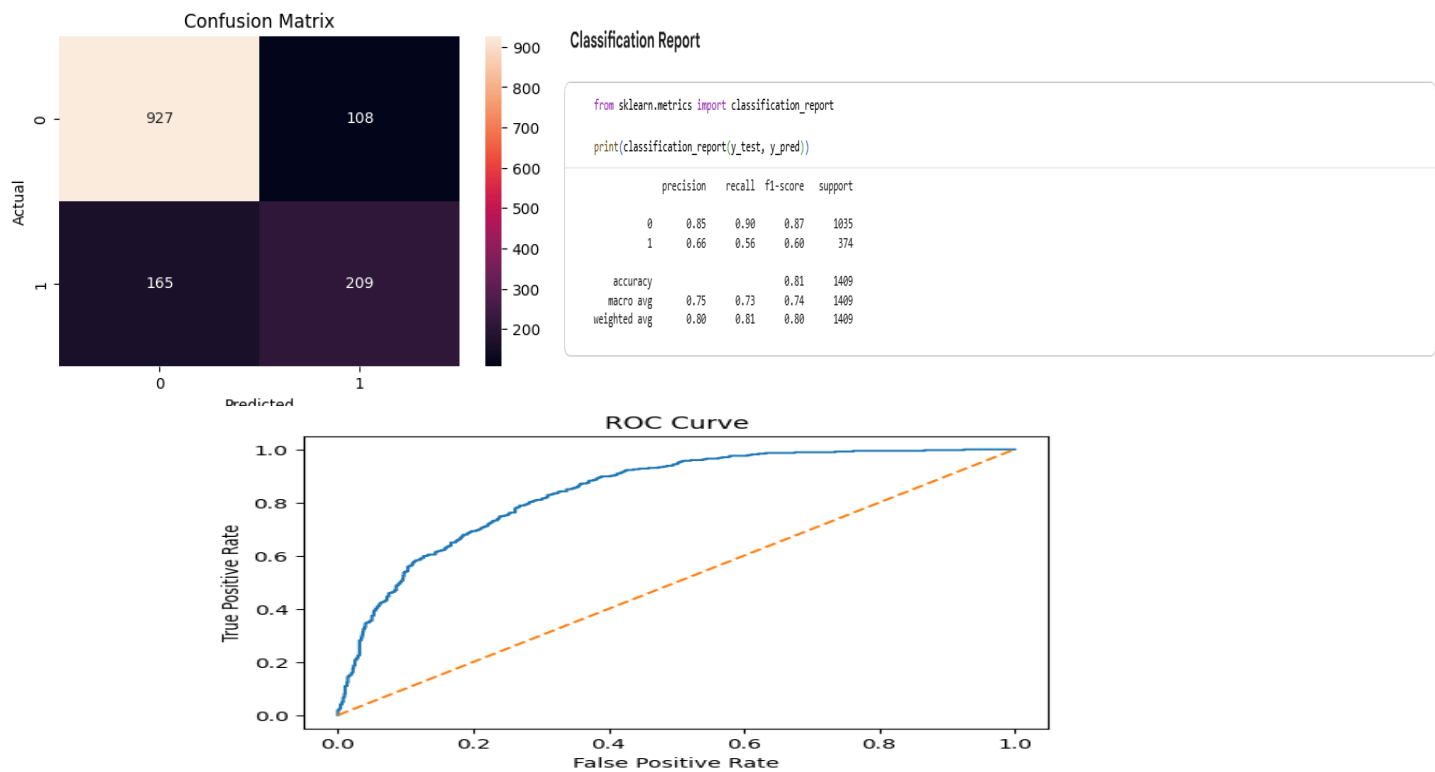
Model Evaluation & Performance Analysis

8.1 Objective

After training multiple machine learning models, the final step was to evaluate their performance using industry-standard classification metrics. The objective was to identify a model that not only achieved good predictive accuracy but also reliably identified customers who were at risk of churn.

8.2 Model Evaluation

The selected **Logistic Regression** model was evaluated using multiple performance metrics, including the **Confusion Matrix**, **Classification Report**, and **ROC-AUC Curve**. These metrics provided a comprehensive assessment of the model’s ability to classify churn and non-churn customers.



8.3 Model Performance Summary

Evaluation Metric	Result
Selected Model	Logistic Regression

Accuracy	80.62%
ROC-AUC Score	84.22%
Precision (Churn)	0.66
Recall (Churn)	0.56
F1-Score (Churn)	0.60

8.4 Business Interpretation

The model achieved **80.62% accuracy** with an **ROC-AUC score of 84.22%**, demonstrating a strong ability to distinguish between customers who are likely to churn and those who are not. While the model performed well overall, recall highlights that some churning customers may still be missed. This insight emphasizes the importance of balancing predictive performance with business priorities when designing customer retention strategies.

Hyperparameter Tuning & Feature Importance

9.1 Objective

After selecting **Logistic Regression** as the final model, hyperparameter tuning was performed to optimize its performance. Feature importance analysis was also conducted to identify the customer attributes that had the greatest influence on churn prediction.

9.2 Hyperparameter Tuning

GridSearchCV was used to identify the optimal combination of model parameters, improving the model’s ability to generalize to unseen data.

Key Python Code

Hyperparameter Tuning

```
LogisticRegression(max_iter=1000)
```

LogisticRegression

LogisticRegression(max_iter=1000)

Import GridSearchCV

```
from sklearn.model_selection import GridSearchCV
```

Define Parameter Grid

```
param_grid_lr = {
    'C': [0.01, 0.1, 1, 10, 100],
    'penalty': ['l1', 'l2'],
    'solver': ['liblinear']
}
```

Run Grid Search

```
grid_lr = GridSearchCV(
    estimator=LogisticRegression(max_iter=1000),
    param_grid=param_grid_lr,
    cv=5,
    scoring='recall',
    n_jobs=-1
)

grid_lr.fit(X_train, y_train)
```

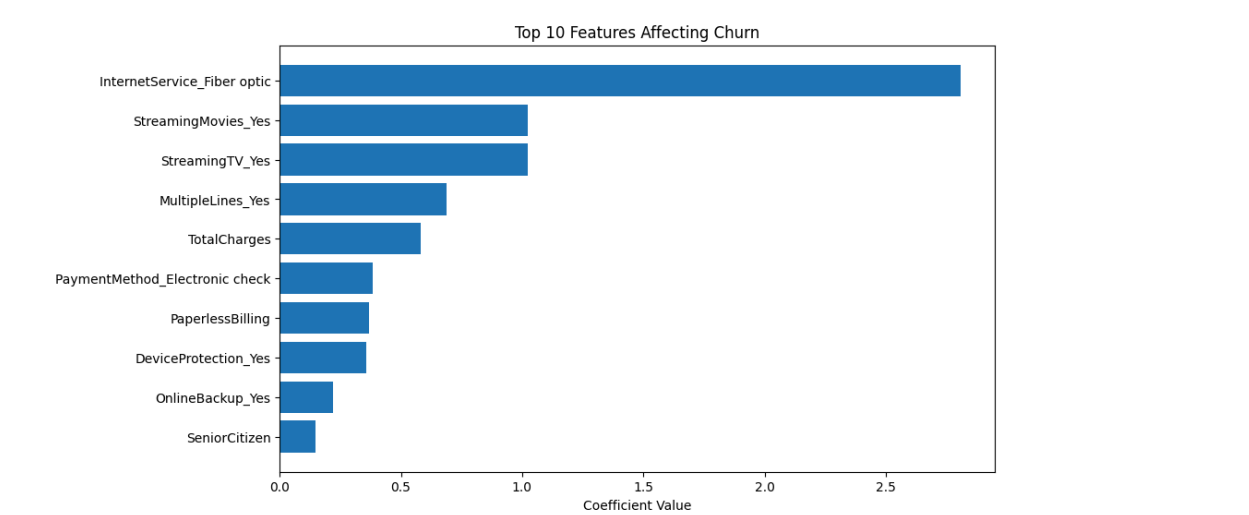
9.3 Optimized Model Performance

Parameter	Best Value
Regularization (C)	100
Penalty	L2
Solver	liblinear
Best Recall Score	0.55

The tuned model improved its ability to identify customers at risk of churn while maintaining stable overall performance.

9.4 Feature Importance

Feature importance analysis highlighted the variables that contributed most to churn prediction, providing valuable business insights beyond the model’s predictions.



Key Business Insights

- Customers with **Month-to-Month contracts** showed a higher likelihood of churn.
- **Short tenure** was strongly associated with customer attrition.
- Customers with **higher monthly charges** were more likely to leave.
- Value-added services such as **Online Security** and **Tech Support** contributed to improved customer retention.

These insights can help businesses prioritize retention campaigns and improve customer engagement strategies.

Business Insights, Learning Outcomes & Conclusion

10.1 Key Business Insights

- Customers with **Month-to-Month contracts** showed the highest likelihood of churn.
- **Short-tenure** customers were more likely to leave the service.
- **Higher monthly charges** were associated with increased churn risk.
- Customers using **Electronic Check** as their payment method exhibited higher churn rates.
- Value-added services such as **Online Security** and **Tech Support** contributed to improved customer retention.

10.2 Business Recommendations

- Encourage customers to switch to long-term contracts through targeted offers.
- Focus retention campaigns on new customers during their initial months.
- Promote value-added services to improve customer engagement.
- Develop personalized retention strategies for high-risk customer segments identified by the model.
- Continuously monitor customer behavior and retrain the model using updated data to maintain prediction accuracy.

10.3 Skills Demonstrated

Python • Pandas • NumPy • Data Cleaning • Exploratory Data Analysis (EDA) • Feature Engineering • Machine Learning • Logistic Regression • Model Evaluation • Hyperparameter Tuning • Predictive Analytics • Scikit-learn • Business Intelligence

10.4 Learning Outcomes

This project strengthened my understanding of the complete Data Science lifecycle, from business problem understanding and data preprocessing to model development, evaluation, and business interpretation. It also enhanced my ability to convert raw data into actionable insights and build practical machine learning solutions for real-world business challenges.

10.5 Future Scope

- Improve performance using advanced ensemble models such as **XGBoost** or **LightGBM**.
- Deploy the solution as an interactive **Streamlit** web application.
- Integrate real-time prediction capabilities using cloud platforms.
- Apply Explainable AI techniques such as **SHAP** for improved model transparency.

10.5 Conclusion

This project demonstrates an end-to-end **Machine Learning** solution for predicting customer churn in the telecommunications industry. By combining data preprocessing, exploratory data analysis, feature engineering, model development, and performance optimization, a reliable predictive model was developed to identify customers at risk of leaving. Beyond achieving strong predictive performance, the project highlights how **Data Science** can support data-driven decision-making, improve customer retention strategies, and deliver measurable business value.